1) September 21 Warm-Up: Profile.java and Runner.java (code and output) - 5 points

```java
public class Profile
{
   private String name;
   private int age;

   public Profile()
   {
       name = "John";
       age = 15;
   }

   public Profile(String name, int age)
   {
       this.name = name;
       this.age = age;
   }

   public void printInfo(int id)
   {
       System.out.print("ID: " + id + " ");
       printVars();
   }

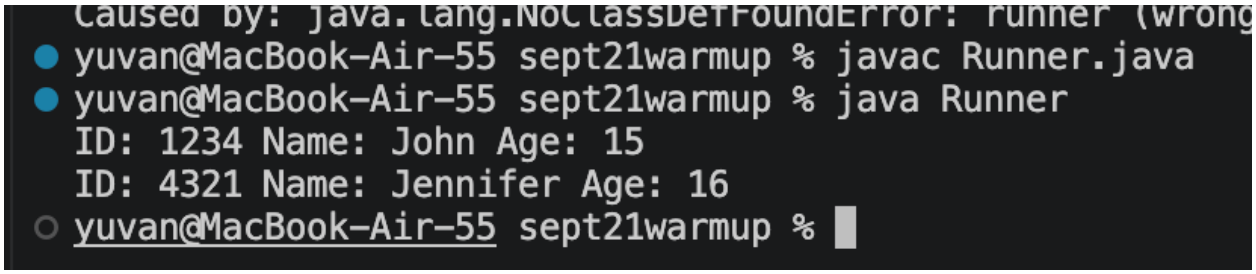
   private void printVars()
   {
       System.out.println("Name: " + name + " Age: " + age);
   }
}
```

```java
public class Runner
{
   public static void main(String[] args)
   {
       Profile p1 = new Profile();
       Profile p2 = new Profile("Jennifer", 16);

       p1.printInfo(1234);
       p2.printInfo(4321);
```

```
    }
}
```

Output:



2) September 22/23 Classwork: UseRadius.java and Runner.java (code and output) - 10 points

```
public class UseRadius {
    private double pi;
    private double radius;
    private double volume;
    private double area;

    public UseRadius(){
        radius = 0;
        pi = 3.14;
        volume = 0;
    }
    public UseRadius(double radius){
        this.radius = radius;
        pi = 3.14;
        volume = 0;

    }
    public void calcVol(){
        volume = (1.33 * Math.pow(radius, 3) * pi);
    }
    public void printVol(){
        calcVol();
        System.out.println("The volume is " + volume);
```

```
   }
   public void printArea(){
       area = pi * Math.pow(radius, 2);
       System.out.println("Area is " + area);

   }
   public void printArea(double radius){
       area = pi * Math.pow(radius, 2);
       System.out.println("Area is " + area);

   }
   public void printVol(double radius, double height){
       volume = height * pi * Math.pow(radius, 2);
       System.out.println("The volume is " + volume);


   }

   /*
   +printArea() : void method that calculates and prints the area of the circle.
   +printArea(double radius) : void method that expects a radius parameter. Calculate
and print the area with radius that is passed in.
   +printVol(double radius, double height) : You already have one printVol() method.
Add this void method that expects a radius and height and calculates the volume of a
cylinder (vol = pi* radius* radius * height).  This method does not call calcVol().


   */

}
```

```
public class Runner {

   public static void main(String[] args){

       UseRadius r1 = new UseRadius();
       UseRadius r2 = new UseRadius(10.0);
```

```
        r1.printVol();
        r2.printVol();
        r1.printArea();
        r2.printArea();
        r1.printArea(5);
        r2.printArea(5);
        r2.printVol(7,10);

    }

    /*
Runner.java - add the following to your code.
Call printArea() and printArea(double) passing in 5 using r1.
Call printArea() and printArea(double) passing in 5 using r2.
Call printVol(double, double) using r2 passing in 7 for radius and 10 for height.
*/
}
```

Output:

```
yuvan@MacBook-Air-55 sept23classwork % javac Runner.java
yuvan@MacBook-Air-55 sept23classwork % java Runner
The volume is 0.0
The volume is 4176.2
Area is 0.0
Area is 314.0
Area is 78.5
Area is 78.5
The volume is 1538.6000000000001
yuvan@MacBook-Air-55 sept23classwork %
```

3) September 24/25 Warm-Up: Rect.java and Runner.java (code) - 5 points

```
public class Rect {
    private int length;
    private int width;
    private int area;

    public Rect(){
        length = 7;
```

```
        width = 7;
    }

    public Rect(int length, int width){
        this.length = length;
        this.width = width;
    }

    public void printArea(){
        calcArea();
        System.out.println("The area is " + area);
    }

    private void calcArea(){
        area = length * width;
    }
}
```

Runner:

```
public class Runner {

    public static void main(String[] args){

        Rect obj1 = new Rect();
        obj1.printArea();

        Rect obj2 = new Rect(5, 2);
        obj2.printArea();

    }
}
```

Output:

```
yuvan@MacBook-Air-55 rectClasswork % javac Runner.java
j%
yuvan@MacBook-Air-55 rectClasswork % java Runner
The area is 49
The area is 10
yuvan@MacBook-Air-55 rectClasswork %
```